Customer Shopping Behavior Analysis

Uncovering insights from 3,900 transactions to drive strategic business decisions

Dataset at a Glance

3,900 Purchases

18 columns spanning demographics, purchase details, and shopping behavior

Key Data Points

- Customer demographics (age, gender, location, subscription)
- Purchase details (category, amount, season, size, color)
- Behavioral metrics (discounts, promo codes, frequency, ratings)

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Data Preparation & Engineering

01

Data Loading & Exploration

Imported dataset using pandas, checked structure with df.info() and summary statistics with .describe()

02

Missing Data Handling

37 missing values in Review Rating column imputed using median ratings by product category

03

Feature Engineering

Created age_group bins and purchase_frequency_days from raw purchase data

04

Database Integration

Connected to PostgreSQL and loaded cleaned DataFrame for SQL analysis

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SQL Analysis: Revenue Insights

Revenue by Gender

Compared total revenue generated by male vs. female customers

High-Spending Discount Users

Identified customers using discounts while spending above average purchase amounts

Revenue by Age Group

Calculated total revenue contribution of each age demographic segment

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SQL Analysis: Product Performance

Top 5 Products by Rating

Products with highest average review ratings across all categories

Discount-Dependent Products

5 products with highest percentage of discounted purchases

Top 3 Products per Category

Most purchased products identified within each product category

SQL Analysis: Customer Segmentation

1

New Customers

First-time purchases identified

2

Returning

Multiple purchases, building loyalty

3

Loyal

High-frequency buyers with strong engagement

Key Comparisons

- Subscribers vs. non-subscribers: average spend and total revenue
- Shipping type comparison: Standard vs. Express purchase amounts
- Repeat buyers: customers with >5 purchases and subscription likelihood

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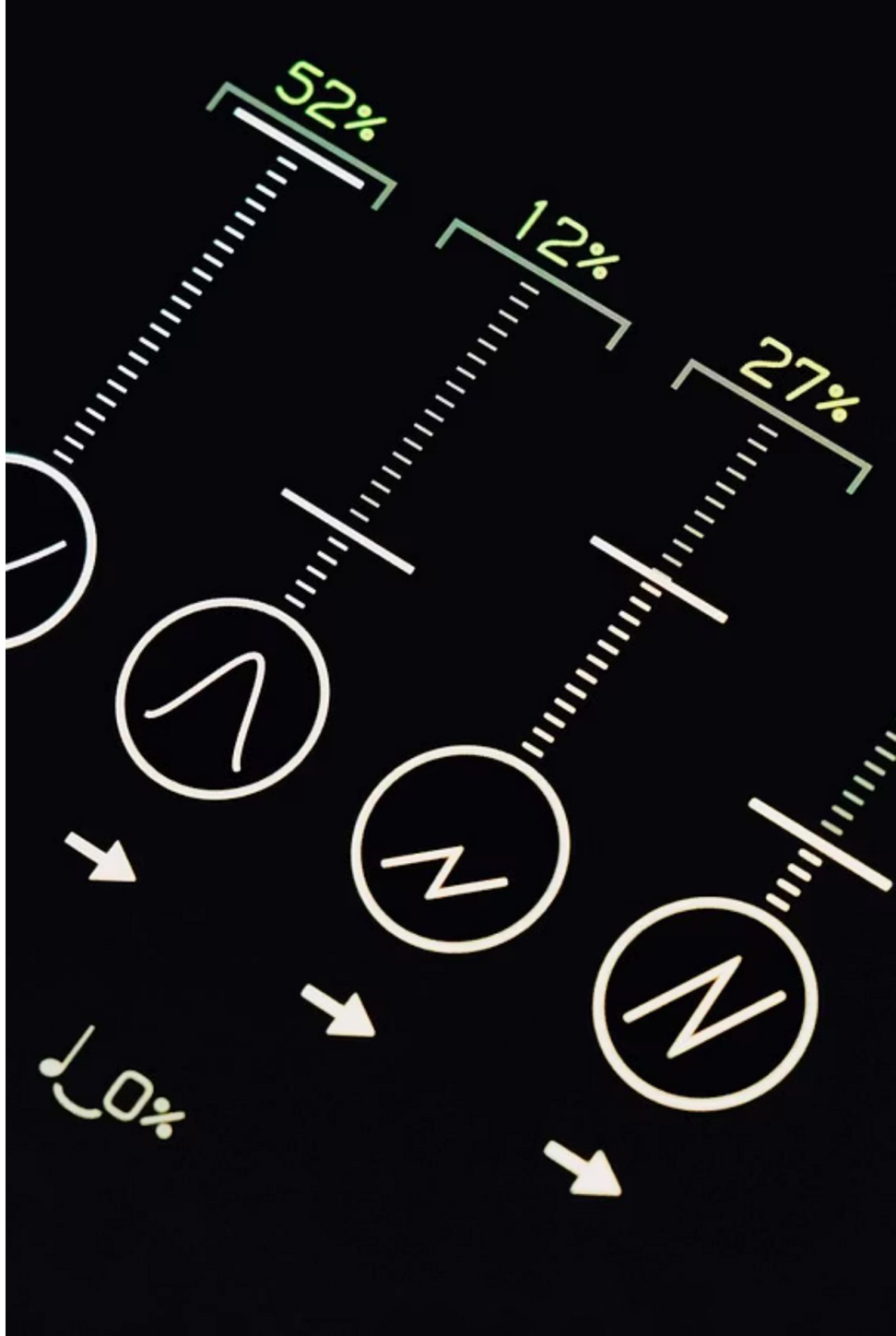
Visual Dashboard in Power BI

Interactive dashboard built to present insights visually for stakeholder review



Revenue by Gender

$212,800

Female Customers

Total revenue generated

$198,500

Male Customers

Total revenue generated

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Category & Discount Analysis

68%

Clothing

Highest discount usage

45%

Home & Garden

Mid-range discount usage

39%

Sports

Lower discount dependency

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Strategic Business Recommendations

Boost Subscriptions

Promote exclusive benefits and incentives for subscriber acquisition

Customer Loyalty Programs

Reward repeat buyers to move them into loyal customer segment

Review Discount Policy

Balance sales boosts with margin control across categories

Product Positioning

Highlight top-rated and best-selling products in marketing campaigns

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users

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